Module 03: Data Visualization

Mohan Babu University - Data Science Digital Library
Comprehensive Study Curriculum - 10 Detailed Chapters

Course Overview:
This module provides an exhaustive study of data visualization, exploratory graphical analysis, perceptual psychology, and visual communication. Students will master visual encoding channels, Matplotlib, Seaborn, interactive dashboards, geospatial cartography, and publication-ready data storytelling.

Curated by Faculty: S.Bosubabu
Department of CSE (Data Science)
Mohan Babu University

Chapter 1: Importance of Visualization & Visual Perception

1. Theoretical Concepts & Visual Theory:
Importance of Visualization & Visual Perception serves as a foundational pillar in effective data communication. Visual encodings map abstract data attributes to graphical features such as position, length, area, angle, and hue. Grounded in Gestalt psychology and Cleveland-McGill perceptual hierarchies, proper visual design optimizes cognitive load and accelerates insight extraction.

2. Methodological Design & Graphical Principles:
Designing graphics requires balancing expressive fidelity with clarity. Key considerations include eliminating visual clutter (chartjunk), maintaining honest aspect ratios, applying perceptually uniform color scales (e.g., Viridis, Plasma), and structuring visual hierarchy.

3. Practical Python Implementation:
Using Python's scientific visualization ecosystem, students implement workflows utilizing Matplotlib for granular control and Seaborn for statistical abstractions. Techniques include multi-axis subplots, faceted grids, kernel density estimations, and categorical distributions.

4. Laboratory Case Studies & Application:
Applied domains include financial market dashboards, epidemiological geospatial heatmaps, high-dimensional dimensionality reduction plots (t-SNE/UMAP), and interactive executive storytelling decks.

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Chapter 2: Human Perception, Color Theory & Visual Encodings

1. Theoretical Concepts & Visual Theory:
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Chapter 3: Introduction to Matplotlib & Figure Architecture

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Chapter 4: Statistical Plots with Seaborn (Distributions & Categorical)

1. Theoretical Concepts & Visual Theory:
Statistical Plots with Seaborn (Distributions & Categorical) serves as a foundational pillar in effective data communication. Visual encodings map abstract data attributes to graphical features such as position, length, area, angle, and hue. Grounded in Gestalt psychology and Cleveland-McGill perceptual hierarchies, proper visual design optimizes cognitive load and accelerates insight extraction.

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Chapter 5: Interactive Dashboards with Plotly & Dash

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Interactive Dashboards with Plotly & Dash serves as a foundational pillar in effective data communication. Visual encodings map abstract data attributes to graphical features such as position, length, area, angle, and hue. Grounded in Gestalt psychology and Cleveland-McGill perceptual hierarchies, proper visual design optimizes cognitive load and accelerates insight extraction.

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Chapter 6: Geospatial Visualization & Choropleth Mapping

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Chapter 7: Time Series & Longitudinal Trend Visualization

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Designing graphics requires balancing expressive fidelity with clarity. Key considerations include eliminating visual clutter (chartjunk), maintaining honest aspect ratios, applying perceptually uniform color scales (e.g., Viridis, Plasma), and structuring visual hierarchy.

3. Practical Python Implementation:

Using Python's scientific visualization ecosystem, students implement workflows utilizing Matplotlib for granular control and Seaborn for statistical abstractions. Techniques include multi-axis subplots, faceted grids, kernel density estimations, and categorical distributions.

4. Laboratory Case Studies & Application:

Applied domains include financial market dashboards, epidemiological geospatial heatmaps, high-dimensional dimensionality reduction plots (t-SNE/UMAP), and interactive executive storytelling decks.

Chapter 8: Avoiding Misleading Charts & Visual Integrity

1. Theoretical Concepts & Visual Theory:
Avoiding Misleading Charts & Visual Integrity serves as a foundational pillar in effective data communication. Visual encodings map abstract data attributes to graphical features such as position, length, area, angle, and hue. Grounded in Gestalt psychology and Cleveland-McGill perceptual hierarchies, proper visual design optimizes cognitive load and accelerates insight extraction.

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3. Practical Python Implementation:
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4. Laboratory Case Studies & Application:
Applied domains include financial market dashboards, epidemiological geospatial heatmaps, high-dimensional dimensionality reduction plots (t-SNE/UMAP), and interactive executive storytelling decks.

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2. Methodological Design & Graphical Principles:
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3. Practical Python Implementation:
Using Python's scientific visualization ecosystem, students implement workflows utilizing Matplotlib for granular control and Seaborn for statistical abstractions. Techniques include multi-axis subplots, faceted grids, kernel density estimations, and categorical distributions.

4. Laboratory Case Studies & Application:
Applied domains include financial market dashboards, epidemiological geospatial heatmaps, high-dimensional dimensionality reduction plots (t-SNE/UMAP), and interactive executive storytelling decks.

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Applied domains include financial market dashboards, epidemiological geospatial heatmaps, high-dimensional dimensionality reduction plots (t-SNE/UMAP), and interactive executive storytelling decks.

1. Theoretical Concepts & Visual Theory:

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2. Methodological Design & Graphical Principles:

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3. Practical Python Implementation:

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4. Laboratory Case Studies & Application:

Applied domains include financial market dashboards, epidemiological geospatial heatmaps, high-dimensional dimensionality reduction plots (t-SNE/UMAP), and interactive executive storytelling decks.

Chapter 9: Storytelling with Data & Editorial Presentation

1. Theoretical Concepts & Visual Theory:
Storytelling with Data & Editorial Presentation serves as a foundational pillar in effective data communication. Visual encodings map abstract data attributes to graphical features such as position, length, area, angle, and hue. Grounded in Gestalt psychology and Cleveland-McGill perceptual hierarchies, proper visual design optimizes cognitive load and accelerates insight extraction.

2. Methodological Design & Graphical Principles:
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3. Practical Python Implementation:
Using Python's scientific visualization ecosystem, students implement workflows utilizing Matplotlib for granular control and Seaborn for statistical abstractions. Techniques include multi-axis subplots, faceted grids, kernel density estimations, and categorical distributions.

4. Laboratory Case Studies & Application:
Applied domains include financial market dashboards, epidemiological geospatial heatmaps, high-dimensional dimensionality reduction plots (t-SNE/UMAP), and interactive executive storytelling decks.

1. Theoretical Concepts & Visual Theory:
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3. Practical Python Implementation:
Using Python's scientific visualization ecosystem, students implement workflows utilizing Matplotlib for granular control and Seaborn for statistical abstractions. Techniques include multi-axis subplots, faceted grids, kernel density estimations, and categorical distributions.

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4. Laboratory Case Studies & Application:

Applied domains include financial market dashboards, epidemiological geospatial heatmaps, high-dimensional dimensionality reduction plots (t-SNE/UMAP), and interactive executive storytelling decks.

Chapter 10: Advanced Multi-Panel Techniques & Publication Graphics

1. Theoretical Concepts & Visual Theory:
Advanced Multi-Panel Techniques & Publication Graphics serves as a foundational pillar in effective data communication. Visual encodings map abstract data attributes to graphical features such as position, length, area, angle, and hue. Grounded in Gestalt psychology and Cleveland-McGill perceptual hierarchies, proper visual design optimizes cognitive load and accelerates insight extraction.

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4. Laboratory Case Studies & Application:
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